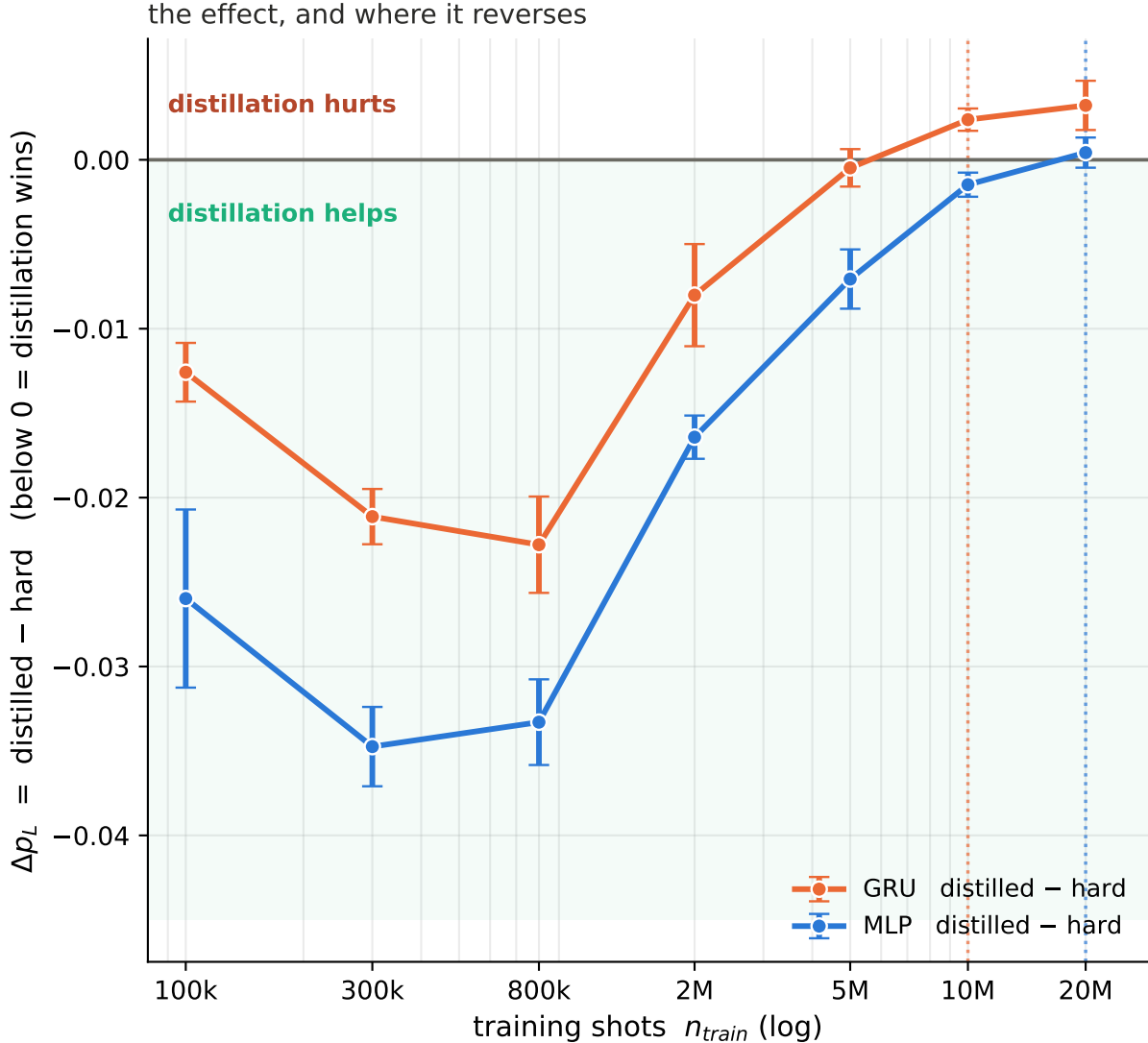
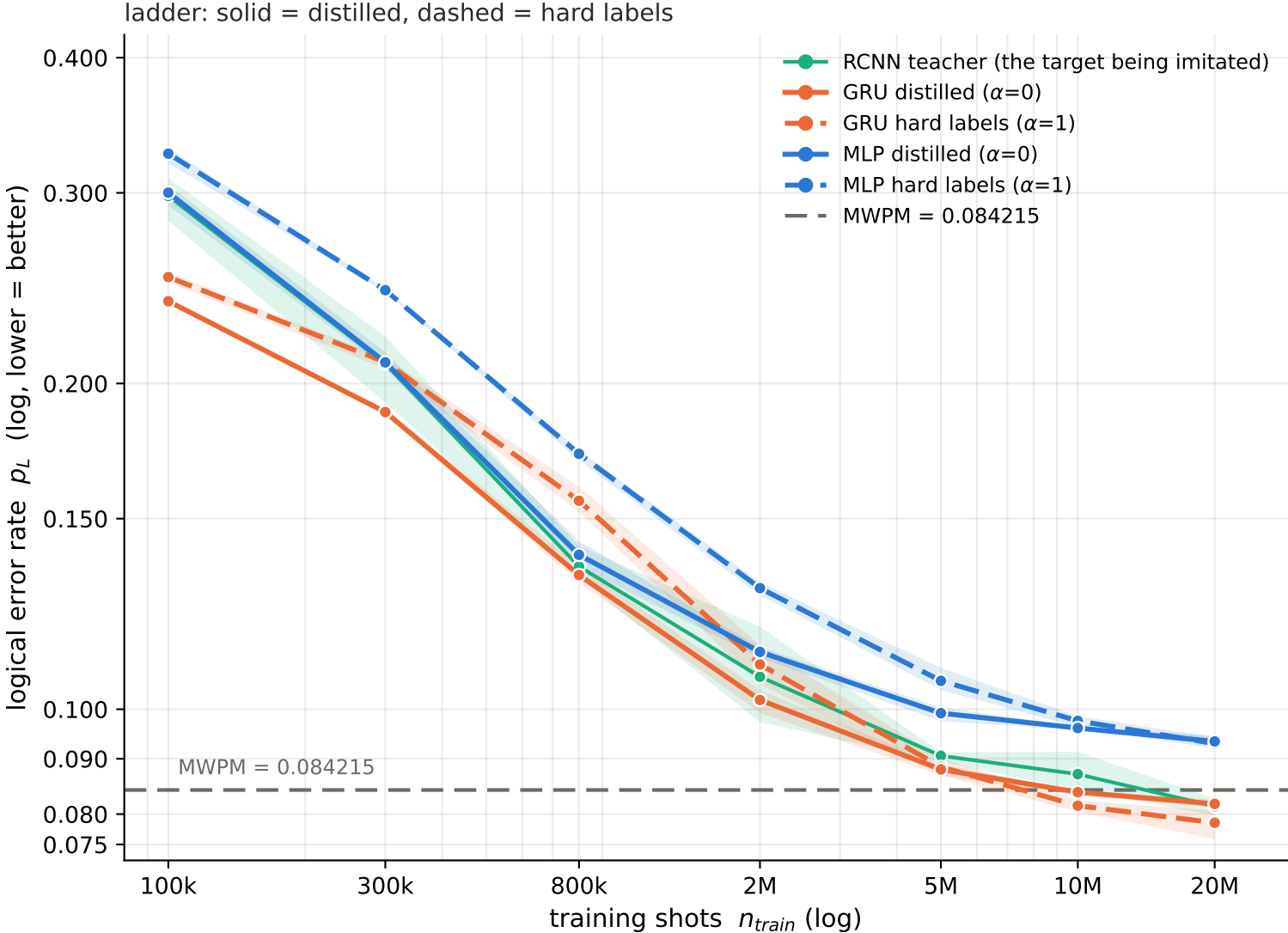


Experiment 9 — distillation vs hard labels at $r=5$

Surface code $d=5$, $p=0.01$, $r=5$ · 200,000-shot validation partition [20.0M, 20.2M) · 3 seeds per rung per arm · band = full range



Teacher for the soft targets: the MEDIAN 20M seed (seed 0, $p_L=0.081585$), not the best, so the students do not inherit an outlier draw. $T=1$, $lr=3e-3$, identical to the hard-label arm. Both arms come from one trainer (train_student.py) differing only in --alpha, and are scored on the same 200,000 shots. The sealed test partition [20.2M, 20.4M) is not read.